PROJECT REPORT

English–Dholuo Medical Translation System

A Bidirectional Text and Speech NLP Pipeline for Kenyan Telemedicine

Team	TRENDLENS
Program	Flatiron School Data Science — DSF-PT14, Phase 5 Capstone
Timeline	2 weeks (14 working days)
Platform	Google Colab (GPU training) + VS Code (development) + GitHub (repo)
Deployment	Hugging Face Hub (model) + Hugging Face Spaces / Gradio (demo app)

1. Executive Summary

Dholuo is spoken by over 4 million people in Kenya yet has almost no presence in digital health tools. Dholuo-speaking patients, particularly elderly patients managing chronic conditions such as diabetes, blood pressure and arthritis are routinely unable to communicate clearly with doctors on telemedicine platforms. Kenya's doctor-to-population ratio is far below WHO guidance, making remote care increasingly important; however, no existing Kenyan telemedicine platform currently offers Dholuo-language support.

This project builds a bidirectional English↔Dholuo translation system that works in both text languages, type in either language and receive a translation back, written. The core trained contribution is a pretrained multilingual translation model fine-tuned with LoRA on 136,625 parallel sentence pairs, supplemented by a translation quality classifier and a cascaded speech pipeline using pretrained ASR and TTS components. The project is framed as a proof-of-concept integration layer for telemedicine platforms like BYON8.

2. Problem Statement

2.1 Background

Kenya faces a severe shortage of healthcare workers, with a doctor-to-population ratio far below WHO-recommended levels. Telemedicine has emerged as a critical solution, with platforms such as BYON8 providing video and chat consultations. However, these platforms operate exclusively in

English or Swahili, excluding millions of Dholuo-speaking Kenyans, many of whom are elderly and monolingual in Dholuo.

2.2 Core Gap

- No publicly available Dholuo medical translation dataset exists.
- No existing telemedicine platform supports Dholuo.
- Elderly patients managing diabetes, arthritis, and maternal health conditions are disproportionately affected by this communication barrier.
- A 2025 NLP-in-Kenya survey confirmed that Dholuo lacks dedicated NLI datasets or deployed models, identifying it as a significant underserved gap.

2.3 Why Dholuo Specifically

Among Kenyan indigenous languages, Dholuo has the most usable NLP infrastructure: POS-tagged data (Kencorpus), FLORES-200/NLLB parallel text coverage, an existing Helsinki-NLP bilingual model, and the newly available DhoNam speech corpus (51+ hours). This makes it the most tractable low-resource Kenyan language for a high-quality, two-week capstone build.

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3. Project Objectives

3.1 Main Objective

To build, evaluate, and deploy a bidirectional English↔Dholuo translation system text fine-tuned on a large parallel corpus and adapted for medical/clinical language, demonstrating feasibility for integration into existing Kenyan telemedicine platforms.

3.2 Specific Objectives

1. Fine-tune a pretrained multilingual translation model (NLLB-200) on the MT560 English–Dholuo parallel corpus using LoRA, and measure improvement over an untuned baseline using BLEU and chrF scores.
2. Build a translation quality classifier and evaluate the fine-tuned model on a small set of conversational phrases as an independent domain-relevance test.
3. Deploy the fine-tuned model to Hugging Face Hub and a Gradio demo to Hugging Face Spaces, producing a permanently accessible, citable artifact.

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4. Target Audience

Dholuo-speaking patients, particularly rural and elderly patients managing chronic conditions. Kenyan telemedicine platforms and providers (e.g. BYON8) seeking to expand language coverage. Healthcare workers who need a reliable communication bridge during remote consultations. Further, expanded to Financial Services, Judiciary, and Agriculture extension services.

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5. Impact

A working translation pipeline would reduce communication barriers between Dholuo-speaking patients and clinicians on telemedicine platforms, improving access to and quality of remote care. Beyond the immediate use case, the project demonstrates a reproducible approach for extending telehealth language support to other low-resource Kenyan languages.

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6. Pre-existing Work

No existing Kenyan telemedicine platform currently offers Dholuo-language support, which represents the core market gap this project addresses. However, the surrounding NLP ecosystem is more mature than for many low-resource languages: Dholuo already has part-of-speech-tagged data (Kencorpus), FLORES-200/NLLB benchmark coverage, and at least one existing bilingual model reducing the risk of working with an unsupported language.

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7. Success Criteria

- Fine-tuned translation should have BLEU score measurably higher than baseline models.
- BLEU score above 30%, and chrF score of 50%
- Final notebook runs top-to-bottom without errors.
- BLEU improvement of at least 5 points over baseline on the MT560 test set.
- Gradio app live on Hugging Face Spaces with a permanent public URL in the README.
- Fine-tuned model published to Hugging Face Hub under the team account, publicly accessible.

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8. Datasets

Dataset — MT560 English–Dholuo Parallel Corpus

Property	Detail
Source	michsethowusu/english-dholuo_sentence-pairs_mt560 (Hugging Face)

Rows	136,625 sentence-aligned English–Dholuo pairs
Features	Two columns: eng (English text) and luo (Dholuo text)
License	Creative Commons Attribution 4.0 International (CC-BY 4.0)
Access	Free download via datasets.load_dataset(), no login required
Format	Parquet, loadable directly into a Hugging Face Dataset object
Scale vs. threshold	136,625 rows is 13x the 10,000-row minimum requirement

Known Limitation

Manual inspection confirms that the MT560 corpus is dominated by religious (Watchtower/Bible-adjacent) text. This is a well-documented pattern in low-resource African language data, as religious organizations have historically funded the majority of translation work into these languages. A model trained only on this corpus will likely produce translations with a formal, devotional register that is poorly suited to casual clinical conversation. This limitation is stated openly in the notebook and is precisely why the medical-term evaluation set and qualitative analysis are essential components of the evaluation strategy.

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9. Data Pipeline

Step 1 Acquisition

Load the MT560 dataset directly via the Hugging Face datasets API. No manual download is required. Load the medical term CSV separately via pandas. Confirm column names, row counts, and data types.

Step 2 Cleaning

- Remove duplicate rows (both columns identical).
- Drop rows with empty or null values in either column.
- Filter out extremely short pairs (under 3 words in either language).
- Filter out extremely long pairs (over 100 words in either language).
- Remove severely misaligned pairs (word ratio below 0.33 or above 3.0).
- Normalize text: strip leading/trailing whitespace, lowercase, standardize punctuation.

Step 3 Exploratory Data Analysis

- Dataset confirmed: 136,625 rows, 0 missing values, perfectly paired.
- Average English sentence: 18.3 words / 90 characters.

- Average Dholuo sentence: 19.1 words / 89 characters (8% expansion factor).
- Only 2.5% of pairs exhibit severe alignment anomalies.
- Religious-text register skew confirmed by manual inspection of sample rows.

Step 4 Train / Validation / Test Split

- 80% training split from MT560 (general model learning).
- 10% validation split (used to select the best checkpoint during fine-tuning).
- 10% test split (held out entirely until final evaluation).
- Medical term set kept fully separate — never included in any of the above splits.

Step 5 — Baseline Evaluation

Run the pretrained dataset on TF-IDF + NN model with no fine-tuning from the test split. Score with BLEU and chrF. These numbers are the reference point against which all subsequent model improvements are measured.

Step 6 — Seq2Seq + LSTM

Tune the pretrained dataset on the Seq2Seq + LSTM model; it is a foundational neural architecture for machine translation. It operates using two main components: The Encoder; processes the input sequence word-by-word, compressing the entire semantic meaning of the source sentence into a single, fixed-size vector. The Decoder takes this final context vector and generates the target translation word-by-word. Monitor validation loss after each epoch. Stop at the best checkpoint.

Step 7 — Seq2Seq + Attention

By adding an Attention Mechanism, the model maps word relations more effectively. It allows the Decoder to dynamically look back at specific words in the source Dholuo sentence while generating each English word. Monitor validation loss after each epoch. Stop at the best checkpoint.

Step 8 — Transformer Model

The Transformer offers a shift in deep learning by completely abandoning recurrence (LSTMs/RNNs) in favor of Self-Attention. By processing entire sequences at once rather than step-by-step, the Transformer achieves: Massive Parallelization; Highly optimized training on modern GPUs, drastically reducing training times; and Multi-Head Attention: The model can simultaneously focus on different parts of a sentence to capture complex syntactic and semantic relationships, regardless of how far apart the words are. Monitor validation loss after each epoch, BLEU and chrF scores.

Step 9 — Hyperparameter tuning with NLLB 200 + LoRA

NLLB-200 is Meta's state-of-the-art 200-language translation model; when paired with LoRA (Low-Rank Adaptation), it demonstrates how to efficiently fine-tune and tune hyperparameters for machine translation. Standard fine-tuning of NLLB-200 is highly resource-intensive. Using LoRA freezes the base model and trains only a tiny fraction (~1%) of the parameters, making hyperparameter sweep runs practical and efficient. Monitor validation loss after each epoch, BLEU and chrF scores.

Step 10 — Model Evaluation

Evaluate the models against key parameters such as BLEU, chrF, accuracy, validation loss, and training time. Compare the performance of each model and choose the best model for translation deployment and model explainability.

Step 11 — Model explainability

For a low-resource Neural Machine Translation (NMT) project, traditional model explainability techniques used in standard classification tasks (like SHAP or LIME) fall short. The best and most practically effective framework for explaining the tuned NLLB-200 model combines Attention Matrix Visualization and Saliency-based Token Attribution.

Step 12 — Project Insights, Recommendations and Conclusions.

Document the project insights, business case, and scalability.

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10. Models

10.1 Architecture Overview

The model architecture is presented and compared against the following parameters, and the best model is applied for deployment.

Model	BLEU	Validation Loss	chrF	Training time
TF-IDF + NN	24.50	-	-	
Seq2Seq + LSTM	12.40	35.564	28.10	0.0
Seq2Seq + Attention	16.80	5.737	33.40	45.0
Transformer Model	9.38	3.832	42.10	133.5
NLLB 200 + LoRA	34.49	1.168	50.74	62.83

10.2 Why These Model Choices

- NLLB-200-distilled-600M: explicitly trained on Dholuo as one of 200 languages, making it the best available starting point for this language pair. The distilled 600M variant runs on a free-tier GPU.

- LoRA: allows fine-tuning with ~1-2% of parameters updated, reducing compute and memory requirements while achieving comparable results to full fine-tuning. Used in production by C-elo Labs for Kikuyu translation.
- Meta MMS: the only publicly available pretrained ASR and TTS model that explicitly lists Dholuo (Luo) as a supported language.
- LSTM Seq2Seq (already completed): served as the first baseline, confirming that training from scratch on this data leads to severe overfitting (validation loss ~11.8), providing evidence-based justification for the pretrained model approach.

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11. Evaluation Strategy

Translation Model Metrics

- BLEU (Bilingual Evaluation Understudy): measures n-gram overlap between model output and human reference. Range 0–100; higher is better. Low-resource language targets: above 15 is strong.
- chrF (Character n-gram F-score): measures character-level overlap. More appropriate than BLEU for morphologically rich languages like Dholuo, where the same root appears in many surface forms.
- Accuracy: overall proportion of correctly classified good/bad translation pairs.
- Training time: the time the model takes to complete the training and learning.

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12. Deployment

The fine-tuned NLLB-200 + LoRA model dataset was deployed on https://huggingface.co/spaces/roy123njuguna/Eng_Luo and made available publicly.

***** To have multiple attempts, kindly create a profile and login into huggingface.***

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13. Tools & Methodologies

Category	Tools
Environment	Kaggle (GPU T4 X2), PyTorch
Data & EDA	Pandas, NumPy, Matplotlib, WordCloud, NLTK

Preprocessing	re, string, NLTK tokenizers
Baseline	TF-IDF+NN and Seq2Seq + LSTM
Deep ML	Seq2Seq + Attention, Transformer Model, and NLLB 200 + LoRA
Deployment	Huggingface